Non-parametric Tests (Lecture 5 and 6) RStudio Examples

#LECTURE 5: Friedman Test#

Class example 1 (ordinal data)

#read in the data set (assuming that a working directory has already been set with setwd(“”)

```
fr.1 <- read.csv("Lecture5_Example1.csv", header = TRUE, sep = ",")
```

#view the data set to make sure that it imported correctly

```
View(fr.1)
```

#friedman test carried out in two different ways:
 # (1): we specify y (numeric/observation column), then the groups and then the blocks
 # (2): we use a special formula a~b|c where a = observations, b = groups/treatment and c = blocks, togehter with the data set

```
friedman.test(y = fr.1$rating, groups = fr.1$manager, blocks = fr.1$applicant) #(1)
friedman.test(formula = rating~manager|applicant, data = fr.1) #(2)
```

#output differs from what we calculated by hand - our test statistic was 10.6125 and 0.01<p-val<0.025
#conclusion remains the same, though - we reject H0

Class example 2 (non-normal quantitative data)

#read in the data set

```
fr.2 <- read.csv("Lecture5_Example2.csv", header = TRUE, sep = ",")
```

#view the data set to make sure that it imported correctly

```
View(fr.2)
```

#friedman test carried out in two different ways:
 # (1): we specify y (numeric/observation column), then the groups/treatment and then the blocks
 # (2): we use a special formula a~b|c where a = observations, b = groups/treatment and c = blocks, togehter with the data set

```
friedman.test(y = fr.2$bid, groups = fr.2$company, blocks = fr.2$plot) #(1)
friedman.test(formula = bid~company|plot, data = fr.2) #(2)
```

#output differs from what we calculated by hand - our test statistic was 15.25 and 0.001<p-val<0.0025
#conclusion remains the same, though - we reject H0

#LECTURE 6: Spearman's Rank Correlation Test#
#read in the data

```
sp <- read.csv("Lecture6_Example.csv")

#check to see that the data imported correctly

View(sp)

#cor.test() with method="spearman"
#specify the two paired samples - x = time and y = mark
#there are ties - need to include exact = FALSE to obtain the approx p-value
#S = sum of squared differences

cor.test(x=sp$time, y=sp$mark, method = 'spearman', exact= FALSE)
```
